Applications: Growth and Decay

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Growth models

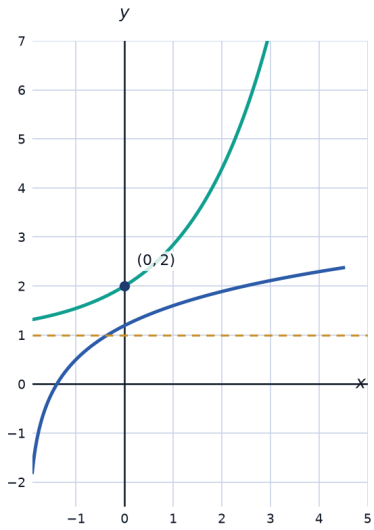
A city's population is modeled by $P(t) = 12000(1.035)^t$, where t is in years.

a State the initial population and the annual growth rate.

b Find the population after 10 years, to the nearest person.

2 [Foundational] • Target: Growth models

An account earns 4% annual interest compounded continuously. How long does it take an investment to double? Give an exact expression and a decimal to two places.



3 [Foundational] • Target: Decay models

Carbon-14 has a half-life of 5730 years: $A(t) = A_0\left(\frac{1}{2}\right)^{t/5730}$.

a What fraction of the original carbon-14 remains after 2000 years? Answer to three decimal places.

b A sample retains 30% of its carbon-14. Estimate its age to the nearest hundred years.

4 [Foundational] • Target: Decay models

A 200 g sample of a radioactive substance decays to 50 g in 12 hours, following $A(t) = 200e^{kt}$.

a Find the decay constant k (4 d.p.).

b Find the half-life of the substance.

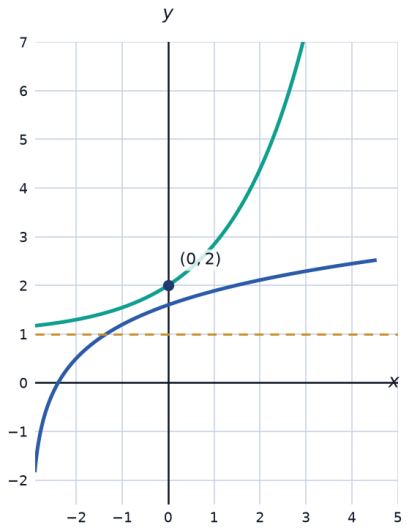
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5 [Foundational] • Target: Decay models

Newton's law of cooling: coffee at $90°\text{C}$ cools in a $20°\text{C}$ room following $T(t) = 20 + 70e^{-0.1t}$ (t in minutes). Find the temperature after 10 minutes, to one decimal place.

6 [Foundational] • Target: Logistic growth (an introduction)

A lake is stocked with fish, and the population is modeled by $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$, where t is in years.



a Find the initial population $P(0)$.

b Find the carrying capacity (the value P approaches as $t \to \infty$).

c Find $P(10)$, to the nearest fish.

7 [Foundational] • Target: Comparing models

Two towns start with 10,000 people. Town A grows at a constant rate of 300 people/year (linear); Town B grows at 3% per year (exponential). Find each town's population after 15 years, to the nearest person, and state which is larger.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

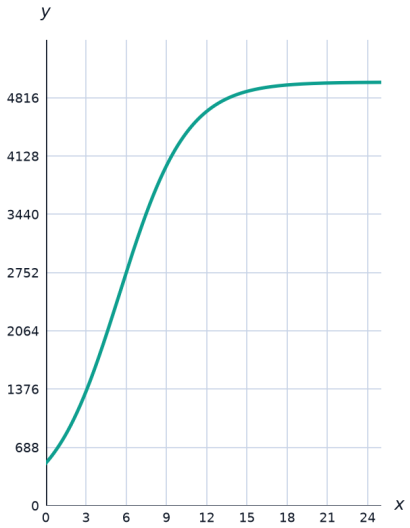
8 [Intermediate] • Target: Comparing models

A radioactive isotope decays from 80 g to 10 g over 3 half-lives. Find the half-life in terms of the decay time, and write the exponential decay model $A(t) = 80e^{kt}$, finding k if the total elapsed time is 15 years (4 d.p.).

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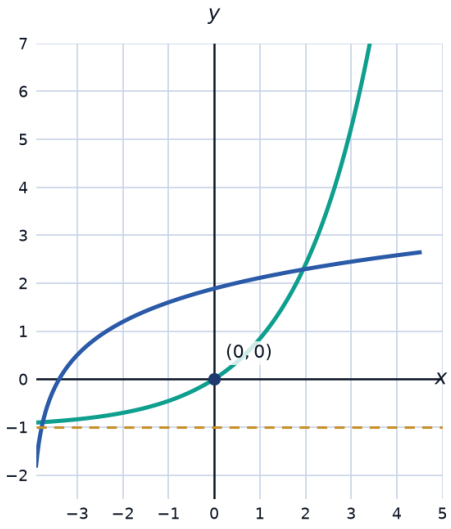
9 [Intermediate] • Target: Visualizing the logistic curve

The fish population model $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$ (from earlier) is graphed below. Use the graph to estimate where the population is growing fastest, then explain why this happens near the inflection point.



10 [Intermediate] • Target: Doubling time and half-life formulas

Derive a general formula for the doubling time T of a quantity growing as $A(t) = A_0 e^{kt}$, in terms of k.



11 [Intermediate] • Target: Doubling time and half-life formulas

Using the doubling-time formula derived above, find the doubling time for a population growing at a continuous rate of $k = 0.025$ per year, to one decimal place.

12 [Intermediate] • Target: Half-life formula derivation

Derive a general formula for the half-life $T_{1/2}$ of a quantity decaying as $A(t) = A_0 e^{kt}$ (with $k < 0$), in terms of k.

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13 [Intermediate] • Target: Half-life formula derivation

A radioactive isotope has decay constant $k = -0.023$ per year. Use the formula above to find its half-life, to one decimal place.

14 [Intermediate] • Target: Combining growth models with rates

A savings account earns continuous interest, growing from 5000 to 5512.50 riyals in 2 years. Find the annual interest rate r using $A = Pe^{rt}$, to two decimal places (as a percentage).

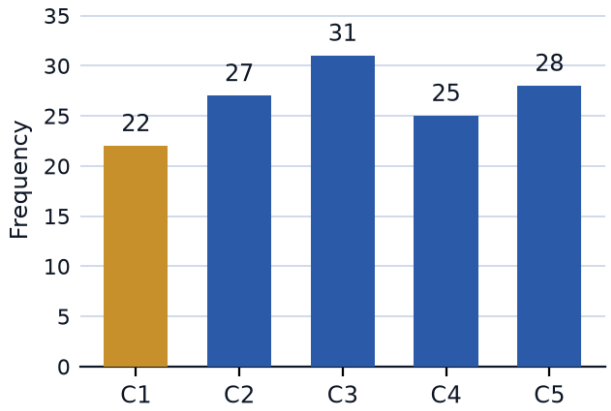
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Half-life formula derivation • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

The half-life of Carbon-14 is 5730 years. (a) Write $A(t)$. (b) How much of 200 g remains after 11 460 years? (c) When is only 10% left?

16 [Challenging] • Target: Combining growth models with rates • Cross-Lesson: 3.03 Sum and Difference Formulas

Level 2 · MediumA table of values for a positive function $g(x)$ over equally spaced inputs $x = 0, 1, 2, 3$ produces outputs $g(0) = 4, g(1) = 12, g(2) = 36, g(3) = 108$. Which function family best models $g(x)$, and what is the constant growth factor per unit increase in x?



17 [Challenging] • Target: Growth models • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

A stone is thrown vertically upwards with speed 15 m/s from the top of a cliff 40 m above sea level. Taking $g = 9.8$ m/s^2 and upwards as positive: (a) find the maximum height reached above the cliff, (b) calculate the time taken for the stone to reach the sea.

18 [Challenging] • Target: Decay models • Cross-Lesson: 4.01 Circles and Parabolas

Find the equation of the normal to the curve $y = \frac{4}{\sqrt{x}} + 2x$ at the point where $x = 4$.

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19 [Challenging] • Target: Logistic growth (an introduction) • Cross-Lesson: 4.02 Ellipses

A Ferris wheel has a diameter of 40 meters and its lowest point is 4 meters above the ground. Passengers board at the lowest point at $t = 0$, and the wheel completes one full revolution every 8 minutes. Write a cosine model $h(t)$ for a passenger's height at time t, and find the times during the first revolution when $h(t) = 34$ meters.

20 [Challenging] • Target: Comparing models • Cross-Lesson: 4.03 Hyperbolas

The temperature θ of a cooling metal rod satisfies $\frac{d\theta}{dt} = -k(\theta - 20)$, where t is time in minutes and k is a constant. Given $\theta = 80$ when $t = 0$ and $\theta = 50$ when $t = 10$, find the value of t when $\theta = 30$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Growth models

The doubling time model is: $A = A_0 \cdot 2^{t/d}$. If $d = 5$, find A/A_0 at $t = 10$.

a. 8

b. 2

c. 4

d. 10

22 [Foundational] • Target: Decay models

For $A = A_0 e^{-kt}$, find the half-life $t_{1/2}$ in terms of k.

a. $\frac{k}{\ln 2}$

b. $\frac{\ln 2}{k}$

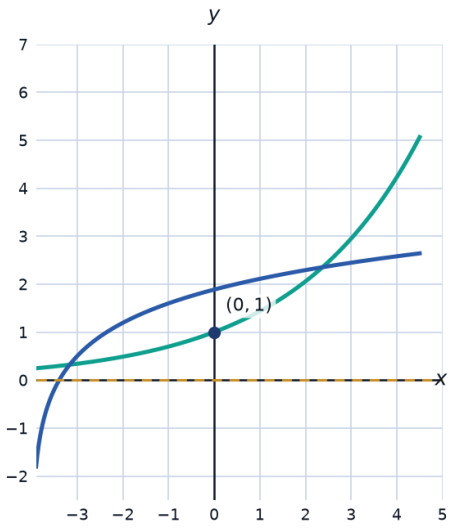
c. $\frac{1}{k}$

d. e^k

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23 [Foundational] • Target: Logistic growth (an introduction)

A committee of 3 people is to be selected from a group of 5 people. The number of ways to form this committee is given by the combination formula ${}^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$. How many different committees can be formed?



a. 10

b. 15

c. 20

d. 60

24 [Foundational] • Target: Comparing models

A sequence is defined by the recursive formula $a_n = 2a_{n-1} + 1$, with the first term $a_1 = 3$. What is the value of the third term, a_3?

a. 7

b. 10

c. 15

d. 21

25 [Foundational] • Target: Visualizing the logistic curve

Consider the arithmetic sequence where the first term $a_1 = 7$ and the common difference $d = -3$. Which of the following is the explicit formula for the nth term of this sequence?

a. $a_n = 7 + 3(n-1)$

b. $a_n = 7 - 3(n-1)$

c. $a_n = 3 - 7(n-1)$

d. $a_n = -3 + 7(n-1)$

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26 [Foundational] • Target: Doubling time and half-life formulas

Prove $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$ using the formula for arithmetic sums.

a. Cannot be proved simply

b. By FTA

c. By induction: base $n = 1$ trivially true; induction step involves algebraic manipulation

d. Use telescoping

27 [Foundational] • Target: Half-life formula derivation

Find the closed form for F_n (Fibonacci) using Binet's formula.

a. $F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}$ where $= 1 + 52, = 1 - 52$

b. $F_n = \phi^{n-1}$

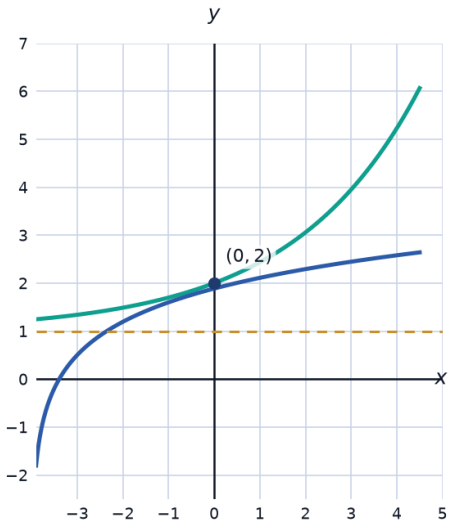
c. $F_n = 2^n/5$

d. $F_n = n\phi$

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28 [Foundational] • Target: Combining growth models with rates

The binomial probability formula is:



a. $P(X = k) = np^k$

b. $P(X = k) = p^k(1 - p)^{n-k}$

c. $P(X = k) = \binom{n}{k}p^k(1 - p)^{n-k}$

d. $P(X = k) = \binom{n}{k}p^n$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Growth models

Derive the formula for the survival function of the exponential distribution.

a. $P(X > t) = \lambda e^{-\lambda t}$

b. $P(X > t) = e^{-\lambda t}$

c. $P(X > t) = 1 - \lambda t$

d. $P(X > t) = 1 - e^{-\lambda t}$

30 [Intermediate] • Target: Decay models

The formula for the sum of n terms of an arithmetic series is:

a. $S_n = \frac{a_1}{1-r}$

b. $S_n = \frac{n}{2}(a_1 + a_n)$

c. $S_n = na_1$

d. $S_n = a_1 \cdot \frac{r^n - 1}{r - 1}$

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31 [Intermediate] • Target: Logistic growth (an introduction)

Formulate the null and alternative hypotheses for the following scenario: "A company claims its average delivery time is 3 days. A customer believes it takes longer."

a. $H_0 : \mu = 10,\ H_1 : \mu < 10$

b. $H_0 : \mu = 70,\ H_1 : \mu > 70$

c. $H_0 : \mu = 500,\ H_1 : \mu < 500$

d. $H_0 : \mu = 3,\ H_1 : \mu > 3$

32 [Intermediate] • Target: Comparing models

Balls are drawn one at a time, without replacement, from a bag of 20 balls; X is the number of red balls drawn. Is X binomially distributed?

a. Yes — but only if $n > 30$

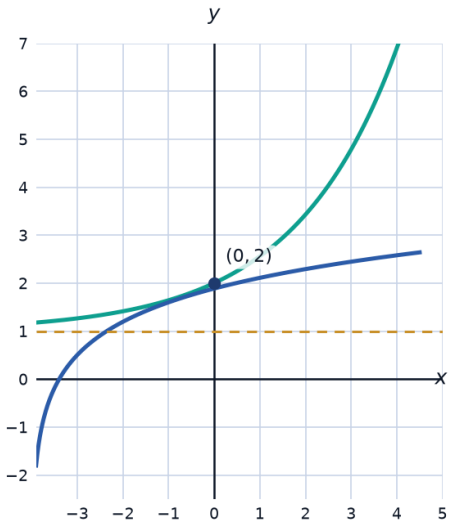
b. Yes — fixed, independent trials with constant success probability

c. No — trials are not independent / probability is not constant

d. No — the number of trials is not fixed

33 [Intermediate] • Target: Visualizing the logistic curve

A curve satisfies $x^2 + y^2 = 25$. Find $\frac{dy}{dx}$ at the point $(3,\ -4)$.



a. $-\frac{3}{4}$

b. $\frac{3}{4}$

c. $\frac{4}{3}$

d. $-\frac{4}{3}$

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34 [Intermediate] • Target: Doubling time and half-life formulas

The formula for the nth term of an arithmetic sequence with first term a and common difference d is:

a. $a_n = a + (n-1)d$

b. $a_n = nd$

c. $a_n = a \cdot d^{n-1}$

d. $a_n = a + nd$

35 [Intermediate] • Target: Half-life formula derivation

From a cumulative frequency curve with $n = 80$, the median is read at cumulative frequency:

a. 20

b. 40

c. 60

d. 80

36 [Intermediate] • Target: Combining growth models with rates

A curve is defined parametrically by $x = 2t^2$, $y = 3t^3$. Find $\frac{dy}{dx}$ at $t = 1$.

a. $\frac{7}{3}$

b. $\frac{9}{4}$

c. $-\frac{9}{4}$

d. $\frac{4}{9}$

37 [Intermediate] • Target: Growth models

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = t^2 - 4t$ m/s. Find when the particle is momentarily at rest.

a. $t = 3$ s (excluding $t = 0$)

b. $t = 5$ s (excluding $t = 0$)

c. $t = 0$ s only

d. $t = 4$ s (excluding $t = 0$)

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Decay models • Cross-Lesson: 4.01 Circles and Parabolas

State Euler's formula and use it to find $e^{i\pi} + 1$.

a. 2

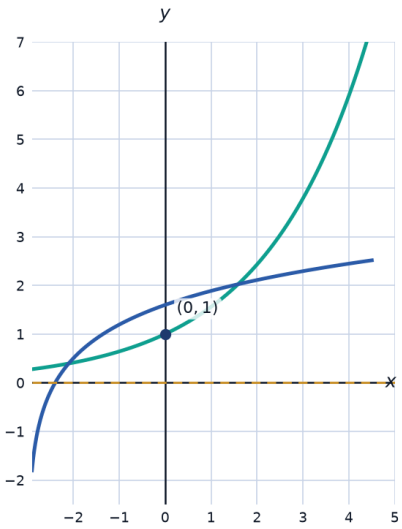
b. e^{π}

c. -2

d. 0 (Euler's identity $e^{i\pi} + 1 = 0$)

39 [Challenging] • Target: Logistic growth (an introduction) • Cross-Lesson: 4.02 Ellipses

A ball is dropped from 10 m and bounces to 80% each time. Total distance (all bounces, both up and down)?



a. 100 m

b. 80 m

c. 50 m

d. 90 m

40 [Challenging] • Target: Comparing models • Cross-Lesson: 4.03 Hyperbolas

Does the point $(4, 0)$ satisfy the implicitly defined curve $x^2 + y^2 = 16$?

a. No, the point does not lie on the curve

b. The equation does not define a curve

c. Yes, the point lies on the curve

d. Only if r is negative

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41 [Challenging] • Target: Visualizing the logistic curve • Cross-Lesson: 5.01 Vectors: Components and Operations

Show that $e^{i\pi} + 1 = 0$ using Euler's formula.

a. $e^{i\pi} = i$

b. $e^{i\pi} = \cos\pi + i\sin\pi = -1 \Rightarrow e^{i\pi} + 1 = 0$

c. $e^{i\pi} = 1$

d. Euler's formula is $e^{ix} = \sin x + i\cos x$

42 [Challenging] • Target: Doubling time and half-life formulas • Cross-Lesson: 5.02 The Dot Product

Find the area enclosed between the curve $y = x$, the x-axis, and the lines $x = 0$ and $x = 3$.

a. 19/2

b. 9

c. 9/2

d. 7/2

43 [Challenging] • Target: Half-life formula derivation • Cross-Lesson: 5.03 Parametric Equations

State and apply Vieta's formulas to find $\sum r_i^2$ if r_1, r_2, r_3 are roots of $x^3 - 4x + 1 = 0$.

a. $\sum r_i^2 = 4$

b. $\sum r_i^2 = 16$

c. $\sum r_i^2 = 1$

d. $\sum r_i^2 = (\sum r_i)^2 - 2\sum_{i<j} r_i r_j = 0 - 2(-4) = 8$

44 [Challenging] • Target: Combining growth models with rates • Cross-Lesson: 6.01 Polar Coordinates and Conversions

The quadratic formula gives: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. For $a = 1, b = -6, c = 8$, find x.

a. $x = -2, -4$

b. $x = 3 \pm 1$

c. $x = 1, 8$

d. $x = 2, 4$

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45 [Challenging] • Target: Growth models • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the sum $1 + 4 + 7 + \cdots + (3n - 2)$ using the formula $S = n(3n - 1)/2$. Verify for $n = 4$.

a. $S_4 = 24$

b. $S_4 = 28$

c. $S_4 = 22$

d. $S_4 = 16$

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